R2.1 How much? The amount of chemical change

Atom economy

Learning outcomes

By the end of this section you should be able to:

- Calculate the atom economy from the stoichiometry of a reaction using the equation provided for you in the DP chemistry data booklet.
- Discuss the inverse relationship between atom economy and wastage in industrial processes.

Percentage yield is a measure of the amount of product obtained in a reaction compared to what should theoretically be produced according to the stoichiometry of the reaction. A high percentage yield (i.e. close to 100%) indicates that the reaction was very efficient at converting reactants into products while a low percentage yield suggests that some of the starting materials may have been wasted or converted into other unwanted byproducts via an alternative reaction.

Recall the recipe for making pancakes which we have revisited throughout this subtopic. The list of ingredients can be written as a chemical reaction (**Figure 1**).

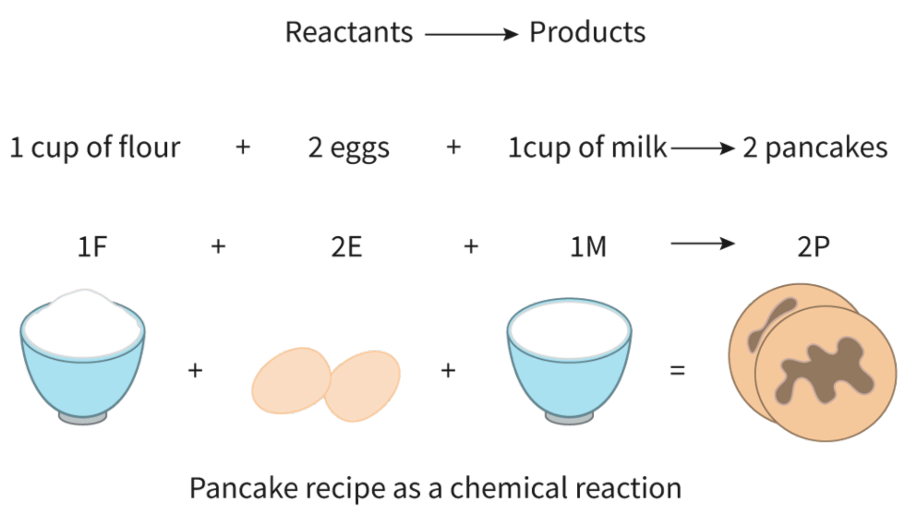

Figure 1. Pancake recipe as a chemical reaction.

More information for figure 1

This diagram illustrates a pancake recipe represented as a chemical reaction. At the top, there's a label connecting 'Reactants' to 'Products.' Below this, the ingredients are lined up along with their respective chemical symbols. "1 cup of flour" is labeled as "1F," "2 eggs" as "2E," and "1 cup of milk" as "1M." These ingredients combine to form the product on the right: "2 pancakes," represented as "2P." Arrows indicate the direction of the reaction from reactants to products. Pictorial representations accompany each ingredient and the pancakes on the far right.

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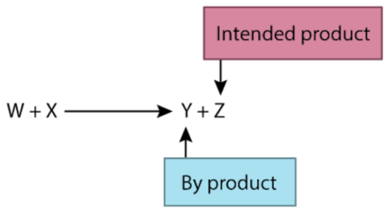
Chefs have worked hard to ensure that the reactants are pure, are mixed properly, are reacting at the right temperature and no spills occur; as a result they receive a very high percentage yield. But is percentage yield the only metric which matters? If one chef puts eggshells in the compost each time and another chef simply puts them in the trash, should this waste be measured?

Egg shells, empty bags of flour and containers of milk are all disposed of after the reaction. What if there was a way to compost the eggshells or recycle the bags and containers? How could this be quantified?

Evaluating efficiency

Chemical reactions are required to produce substances that do not naturally occur, such as medications, food additives or synthetic polymers. In the synthesis of these types of compounds, the desired product is the focus of the reaction, however there might be other substances formed, known as by-products. For example, a chemist who wants to produce a molecule Z via a reaction between W and X, may also produce molecule Y as a by-product (**Figure 2)**.



By-products may also be created in a chemical reaction

Figure 2. By-products may also be created in a chemical reaction.

More information for figure 2

The image is a flowchart showing the process of a chemical reaction that leads to the formation of an intended product and a by-product. The flowchart consists of two labeled rectangles connected by arrows. The first rectangle at the top is labeled "Intended product," indicating the primary goal of the chemical reaction. Below it, an arrow points downward to the second rectangle labeled "By-product," representing additional substances formed during the reaction. This flowchart visually depicts the concept that chemical reactions can yield both desired and unintended products.

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By-products might be useful substances that could be collected and used in another application, or they might serve as waste, causing financial and environmental costs and a challenge for disposal. While percentage yield measures the successful conversion of reactants into the products of the intended reaction, the efficiency of the reaction can also be tracked by another metric: atom economy. This is a measure of the reaction efficiency with respect to the total mass of reactant atoms which end up as desired products, represented as a percentage.

Atom economy is calculated using the total molar masses of reactants and the molar mass of only the desired product:

$$\%\ \text{atom economy} = \frac{\text{Molar mass of desired product}}{\text{Molar mass of all reactants}} \times 100$$

The ideal scenario would be if the atom economy for a reaction was 100%, indicating that all reactants are used to make the desired product with no by-product waste. For two different reactions that make the same desired product using different reactions, the choice where the atom economy is closer to 100% would be the more efficient choice from a green chemistry perspective. The equation is provided for you in section 1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) of the DP chemistry data booklet.

To calculate atom economy you should:

1. Calculate the molar mass of the desired product only (including multiplying by the coefficient).
2. Calculate the total molar masses for all the reactants.
3. Substitute values into the equation.

Worked example 1

Iron can be extracted from its ore in the following reaction:

$$3CO(g) + Fe_2O_3(s) \rightarrow 3CO_2(g) + 2Fe(s)$$

Calculate the atom economy of the reaction.

Solution steps	Calculations
Step 1: Calculate M of the desired product only.	$\text{Desired product} = \text{iron}; M = 55.85 \text{ g r}$ $\text{Total } M = 2\times55.85$ $= 111.70$ (As you are calculating the total M you s coefficient.)
Step 2: Calculate total M for all the reactants.	$(3\times M \text{ C}) + (3\times M \text{ O}) + (2\times M \text{ Fe})$ $(3\times12.01) + (3\times16.00) + (2\times55.85)$ 243.73
Step 3: Substitute values into the equation.	$\text{Atom economy} = \frac{M \text{ of desired pr}}{\text{Total } M \text{ of all rea}}$ $= \frac{111.70}{243.73} \times 100\%$ $= 45.83\%$

Using the law of conservation of mass we can deduce that no atoms are gained or lost in a chemical reaction. However, some of the atoms may not end up in the desired product, and instead form a by-product. The more by-products you have, the greater the likelihood that you will have a low percentage atom economy.

In the example above, iron is the desired product and carbon dioxide is produced as a by-product, and yet less than one-third of the products are iron according to this metric. You will recall from section R1.3.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fossil-fuels-id-45276/) that there is a significant relationship between carbon dioxide levels and global warming. This may encourage the industries which are using this reaction to extract iron to explore other reactions which have a greater atom economy and which reduce harmful waste.

Aspect: Global impact of science

The atom economy and the percentage yield both give important information about the ‘efficiency’ of a chemical process. Think about what else might be needed for a reaction to take place. For example, the energy consumption required in reactions is not measured in either metric; a factor which can have significant global environmental implications. What other metrics do you think might need to be considered?

Atom economy and the amount of waste produced have an inverse relationship. As the waste increases, the percentage atom economy decreases. If a reaction has an atom economy of 31.43%, then 68.57% of the products are waste products.

Practical skills

Tool 1: Experimental techniques — Measuring variables

The ability to identify relationships between variables is an important skill when processing experimental data. Inverse proportionality means that as one variable increases, the other variable decreases, and vice versa. They essentially have an opposite relationship, like when you increase the driving speed, the time it takes to travel that distance decreases.

But what if it was possible to find uses for the by-products? For example, the carbon dioxide produced in the above reaction could be sold for use in fire extinguishers or in carbonated beverages. This has both economic and environmental benefits for the scientist carrying out the reaction.

Suggest what the atom economy would be for a reaction which only produces one product?

Worked example 2

Sodium chloride, table salt, has many uses in everyday life. There are two reactions which can produce sodium chloride:

Reaction 1

$$NaHCO_3(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

Reaction 2

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

Use atom economy to determine which reaction is the more efficient reaction.

Reaction 1

Solution steps	Calculations
Step 1: Calculate the M of the desired product only.	$\text{Desired product} = \text{NaCl}; M = 58.44\ \text{g mol}^{-1}$ $\text{Total } M = 1 \times 58.44 = 58.44$ $(\text{Coefficient of NaCl is } 1)$
Step 2: Calculate total M for all the reactants.	$(1 \times M\ \text{Na}) + (1 \times M\ \text{H}) + (1 \times M\ \text{C}) + (3 \times M$
Step 3: Substitute values into the equation.	$\text{Atom economy} = \frac{M \text{ of desired product}}{\text{Total } M \text{ of all reactants}}$ $= \frac{58.44}{120.47} \times 100\%$ $= 48.51\%$

Reaction 2

Solution steps	Calculations
Step 1: Calculate M of the desired product only.	$\text{Desired product} = \text{NaCl}; M = 58.44\ \text{g mol}^{-}$ $\text{Total } M = 1 \times 58.44 = 58.44$ $(\text{Coefficient of NaCl is } 1)$
Step 2: Calculate total M for all the reactants.	All reactants: $(1 \times M\ \text{Na}) + (1 \times M\ \text{O}) + (1 \times M\ \text{H}) + (1 \times M$
Step 3: Substitute values into the equation.	$\text{Atom economy} = \frac{M \text{ of desired product}}{\text{Total } M \text{ of all reactants}}$ $= \frac{58.44}{76.46} \times 100\%$ $= 76.43\%$

Solution steps	Calculations
	The scientists should choose reaction 2 as it is an atom economy of 76.43% compared to rea

A high atom economy indicates that the reaction was efficient at using up all the reactants and producing minimal waste, while a low atom economy suggests that some of the reactants were not used effectively, that unwanted by-products were formed instead or that there was a high amount of waste. This can be helpful in industry when scientists may need to decide how to produce a particular substance.

Theory of Knowledge

The concept of atom economy was first published by Barry Trost in 1991. Industrial chemists are constantly looking at ways to improve the cost and efficiency of making materials at a large scale. While local governments may address regulations for chemical waste, to what extent are chemists responsible for developing new reaction pathways with high atom economies?

Atom economy and green chemistry

Atom economy is one of the key principles of Green Chemistry; a field of science which focuses on creating chemical processes and products that are sustainable and environmentally friendly. It aims to reduce the negative impact of chemistry on our planet and to promote sustainable practices. This often involves reducing the amount of waste products or non-toxic chemicals, simpler reaction pathways and improved efficiency.

Making connections

Recall the impact of a catalyst on the rate of reaction based on your knowledge from subtopic R 2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43127/) — catalysts alter the speed of a chemical reaction by providing an alternate pathway for the reaction which has a lower activation energy. How do you think this could be beneficial as one of green chemistry's practices?

You will now have an opportunity to consider the importance of different metrics in measuring reaction efficiency through the format of a debate.

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication skills — Practising active listening skills
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group activity

This activity is based on a debate and so your class will need to be divided into two or more teams. Depending on the size of the class, teams in this debate may be composed of 3-5 students.

Instructions:

1. Choose a team captain. This should be someone who is organised, confident and good at communicating. Their role is to keep the team focused and on track during the debate.
2. Allow 15 minutes research time to prepare the arguments for your team and practice presenting them to your team.
3. The debate should follow the format: Each speaker will have no more than with 90 seconds in total and the team arguing in favour will begin. The opposing team should take responsibility for running the timer for each speaker. Remember to practise good sportsmanship; this is a debate, not a fight!

The motion of the debate is: 'Atom economy is a more important metric than percentage yield'.